Fiscal Year:	FY 2019	**Task Last Updated:**	FY 01/02/2019
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Round Robin - Thermophysical Property Measurement		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	04/01/2017	**End Date:**	03/31/2022
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Krivilev, Mikhail Ph.D. (Co-PI: Udmurt State University Izhev, Russia)		
Grant/Contract No.:	NNX17AH41G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Researchers go to space to conduct material property measurements that are required to model terrestrial manufacturing processes to provide higher performance and discover new materials to make people's lives better around the world. We justify the expense of doing tests in microgravity by concentrating on those materials which are impossible to evaluate on Earth or have properties that can only be measured with higher precision when gravitational accelerations are minimized. We know we get better results but we do not yet know how much better. The JAXA (Japanese Aerospace Exploration Agency) Electrostatic Levitation Furnace (ELF) was designed to process high temperature oxides but it can be readily adapted to allow testing of industrially important metals and alloys as well. By selecting materials that are already being processed using another technique, the European Space Agency (ESA) Electromagnetic Levitation facility on the International Space Station (ISS), these two facilities can be compared so that researchers know their relative strengths such that they may complement each other to maximize science output from space tests. Our objectives are to conduct space experiments to compare to ground results on a baseline material set and to model the influence of gravity on measurement techniques by various research groups across the world. Our approach is to employ Round Robin testing by multiple researchers on the same material using different techniques, on different facilities, with different analysis protocols to compare results. The significance of this work is that we will then know the relative increase in measurement fidelity and which material works best in which facility to efficiently use the limited available space sciences resources such as launch cost, crew time, and telemetry bandwidth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This program leverages the existing ESA International Topical Team organization to develop a strategy to compare the performance of current flight and ground facilities investigating thermophysical property measurement using levitation techniques. New experiments are proposed for the JAXA-ELF facility on the ISS to complement other ongoing investigations that use both the ISS-based MSL-EML and ground-based facilities. We will efficiently use available resources, both fiscal and crew time, by selecting materials compatible with those previously prioritized by other existing research groups. In doing so, we will expand the capabilities of space electrostatic levitation processing to allow for investigation of additional thermophysical material properties and manifesting new material classes across microgravity platforms. The key scientific questions we address are: (1) How do we control and minimize gravity-induced systematic error in thermophysical property measurement and does theory explain the observed deviations in results? (2) For which material classes are space-based platforms delivering thermophysical property measurement results with higher accuracy and/or precision than ground-based facilities? Thermophysical property testing in microgravity provides enhanced fidelity of results purportedly due to three main effects. First, without strong gravitational accelerations and with reduced levitation forces a more spherical sample allows for better analysis of experimental behavior. Second, better control of convection in space results in higher measurement precision. Third, sedimentation and buoyancy induced segregation are eliminated in reduced gravity. But how much better do we do in space as compared to ground? The obvious solution is to quantify variability across platforms – both in space and on ground.
Task Progress:	Highlights of Work Accomplished During Grant Period Jan 2018 - Dec 2018. • Initiated preparation of Science Requirements Document (SRD) as the first NASA-Principal Investigator (PI) to conduct flight experiments using the JAXA-ELF on the ISS • Conducted flight sample fabrication prototype tests with Michael SanSoucie at NASA-MSFC (Marshall Space Flight Center) ESL (electrostatic levitation) facility • Hosted flight preliminary flight definition discussions at Tufts University with Dr. Chihiro Koyama of JAXA and Michael SanSoucie of NASA-MSFC • Attended 2018 20th Properties for Materials Science at High Temperature Symposium as session chair and organizer June 27, NIST (National Institute of Standards & Technology)/Univ. Colorado, Boulder CO • Attended ASGSR (American Society for Gravitational and Space Research) 2018 Annual Conference, November 1, Bethesda MD, as symposium organizer and presenter • Conducted ESL density testing on Platinum with visit to NASA/MSFC by Jannatun Nawer (PhD) and Sarah Carty (BS) December 2018 • Honored to be elected President of the National Professional Society ASGSR (American Society for Gravitational and Space Research) with 3-year term on Board of Governors 2018-2021.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Xiao X, Hyers RW, Wunderlich RK, Fecht H-J, Matson DM. "Deformation induced frequency shifts of oscillating droplets during molten metal surface tension measurement." Applied Physics Letters. 2018 Jul;113(1):011903. https://doi.org/10.1063/1.5039336 , Jul-2018
Awards	Matson DM. "Elected President of ASGSR (American Society for Gravitational and Space Research) with 3-year term on Board of Governors 2018-2021, November 2018." Nov-2018